CNN Activation & Gradient Clipping Ablation Report
CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh

ByzFL SNN Benchmark

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Contents

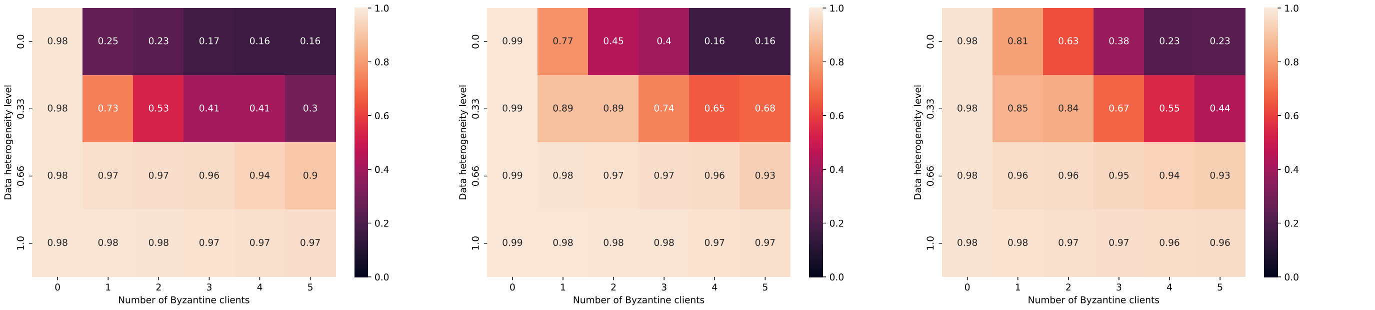
1 Introduction

This report compares three CNN configurations to isolate the effect of gradient saturation/clipping on Byzantine robustness:

1. **CNN ReLU (No Clip)**: Standard CNN with ReLU activation, no gradient clipping ($LR = 0.1$).
2. **CNN ReLU (Grad Clipped)**: Same CNN with ReLU + explicit gradient norm clipping.
3. **CNN Tanh**: Same CNN architecture with Tanh activation (natural gradient saturation).

Note: CNN Clipped results are currently available only for the "best aggregator" heatmaps and the Geometric Median aggregator. Other aggregator sections show CNN ReLU vs CNN Tanh only.

2 Best Overall Performance (Worst-Case Across Attacks)

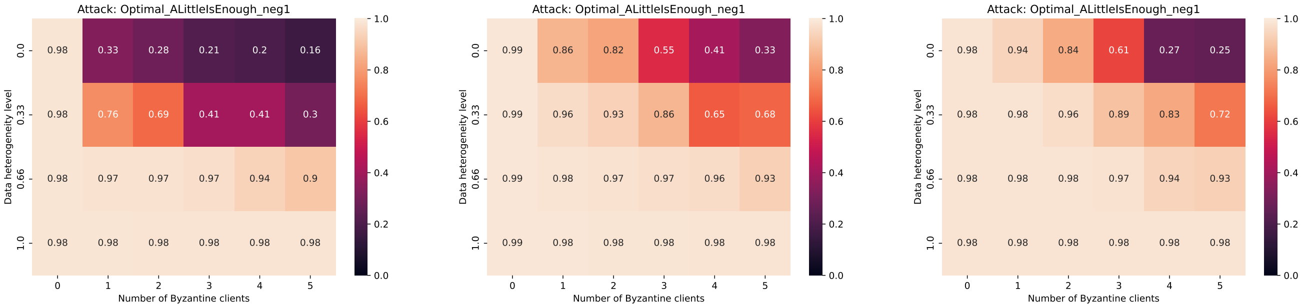


(a) CNN ReLU (No Clip) (b) CNN ReLU (Grad Clipped) (c) CNN Tanh

Figure 1: Best overall (worst-case across attacks): CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh.

3 Best Aggregator under Specific Attacks

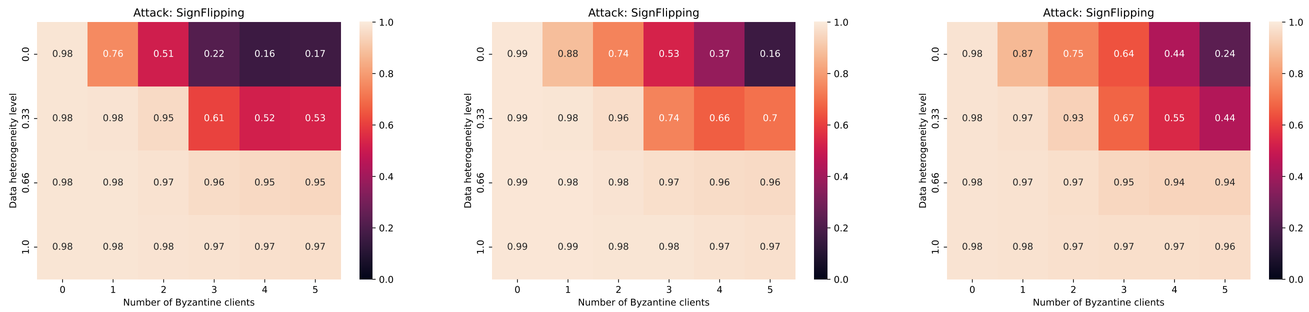
3.1 Optimal ALIE Attack



(a) CNN ReLU (No Clip) (b) CNN ReLU (Grad Clipped) (c) CNN Tanh

Figure 2: Best under OptALIE: CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh.

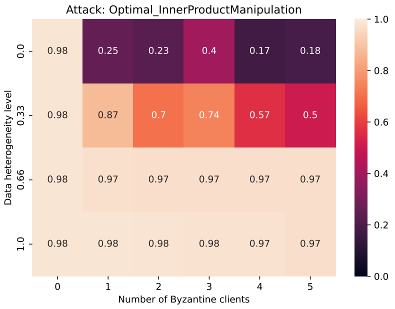
3.2 Sign Flipping Attack



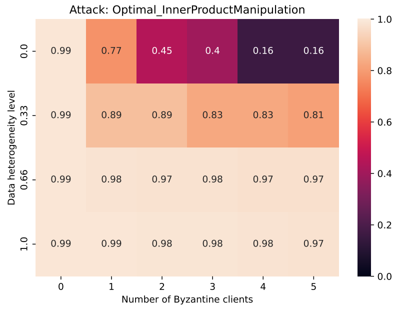
(a) CNN ReLU (No Clip) (b) CNN ReLU (Grad Clipped) (c) CNN Tanh

Figure 3: Best under Sign Flipping: CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh.

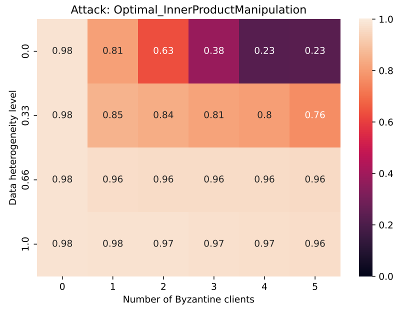
3.3 Optimal IPM Attack



(a) CNN ReLU (No Clip)



(b) CNN ReLU (Grad Clipped)



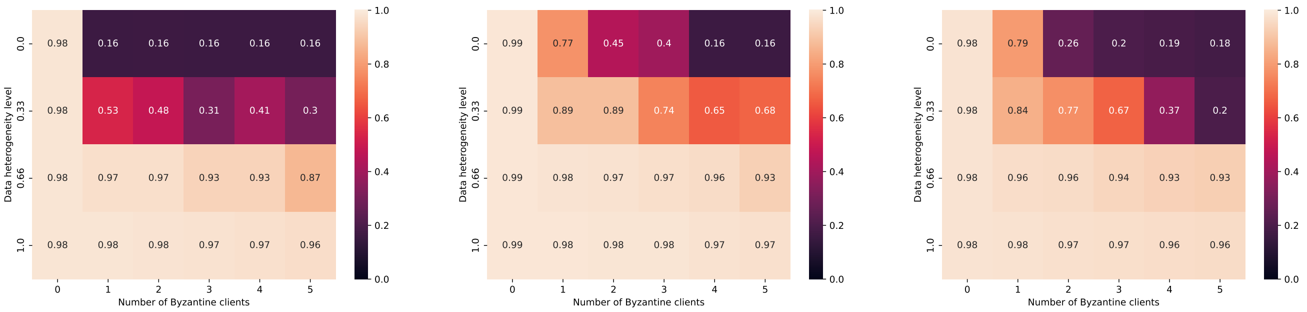
(c) CNN Tanh

Figure 4: Best under OptIPM: CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh.

4 Per-Aggregator Results

4.1 Geometric Median (GM) — All 3 Configs

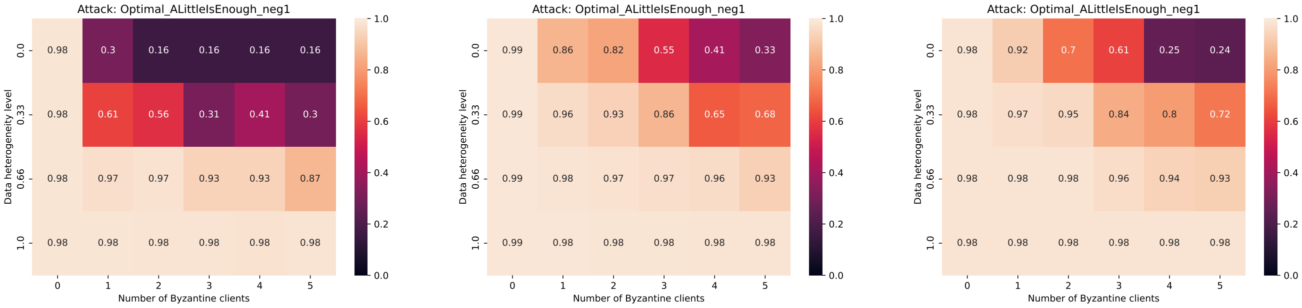
4.1.1 GM under Worst-Case



(a) CNN ReLU (No Clip) (b) CNN ReLU (Grad Clipped) (c) CNN Tanh

Figure 5: Geometric Median under Worst-Case: CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh.

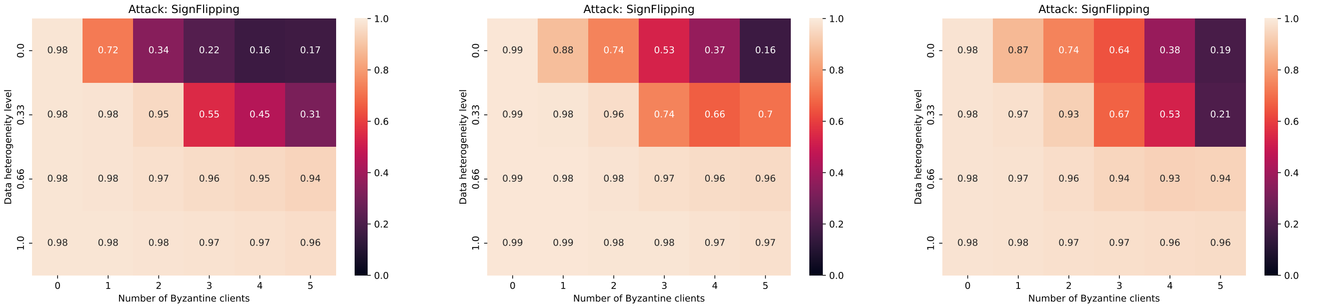
4.1.2 GM under Optimal ALIE



(a) CNN ReLU (No Clip) (b) CNN ReLU (Grad Clipped) (c) CNN Tanh

Figure 6: Geometric Median under Optimal ALIE: CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh.

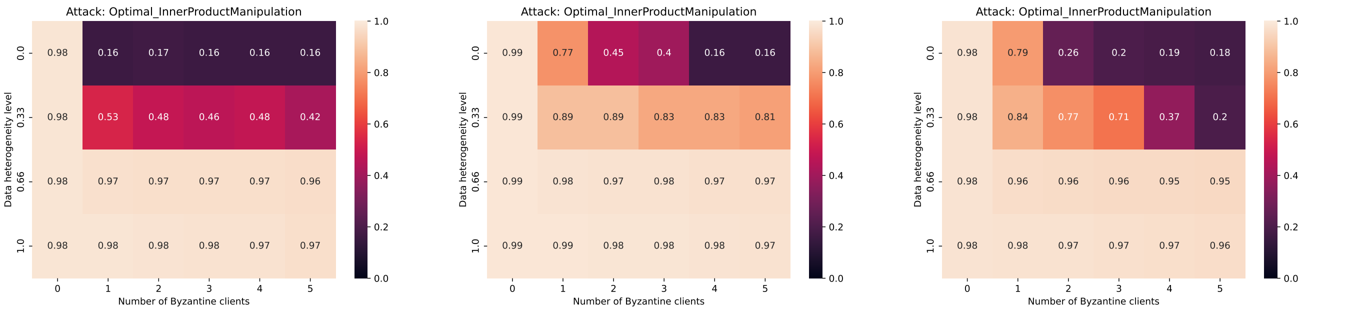
4.1.3 GM under Sign Flipping



(a) CNN ReLU (No Clip) (b) CNN ReLU (Grad Clipped) (c) CNN Tanh

Figure 7: Geometric Median under Sign Flipping: CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh.

4.1.4 GM under Optimal IPM

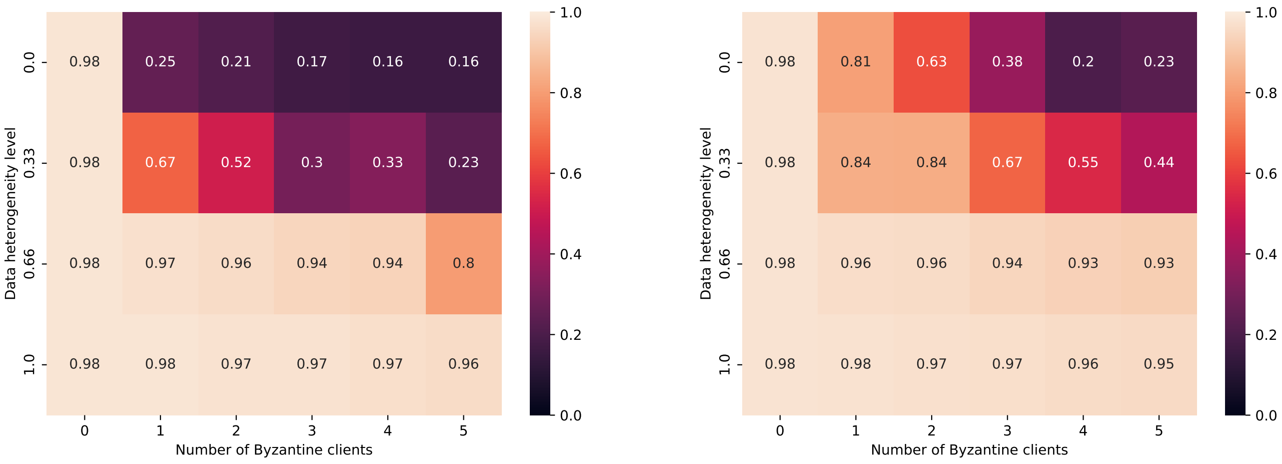


(a) CNN ReLU (No Clip) (b) CNN ReLU (Grad Clipped) (c) CNN Tanh

Figure 8: Geometric Median under Optimal IPM: CNN ReLU (No Clip) vs. CNN ReLU (Clipped) vs. CNN Tanh.

4.2 Centered Clipping (CC) — NoClip vs Tanh

4.2.1 CC under Worst-Case

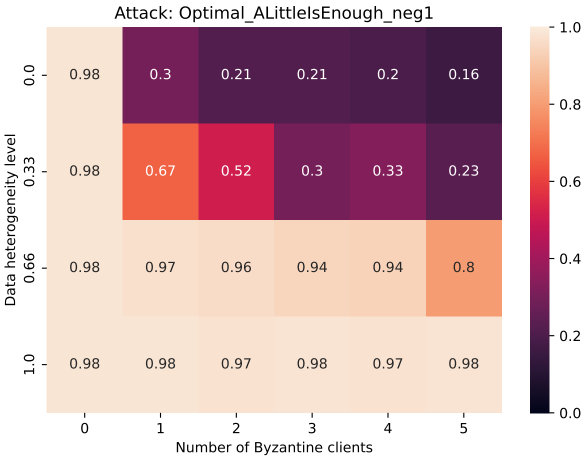


(a) CNN ReLU (No Clip)

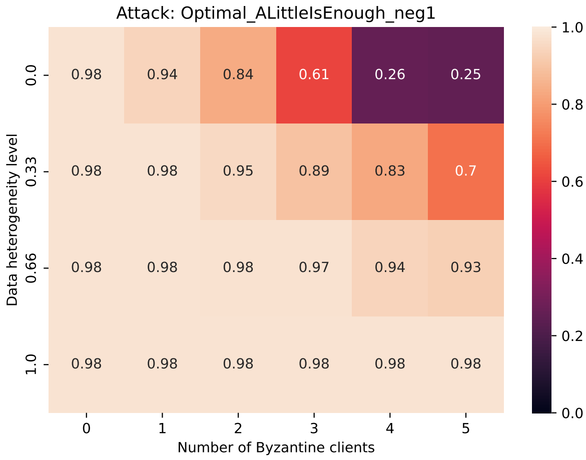
(b) CNN Tanh

Figure 9: Centered Clipping under Worst-Case: CNN ReLU (No Clip) vs. CNN Tanh.

4.2.2 CC under Optimal ALIE



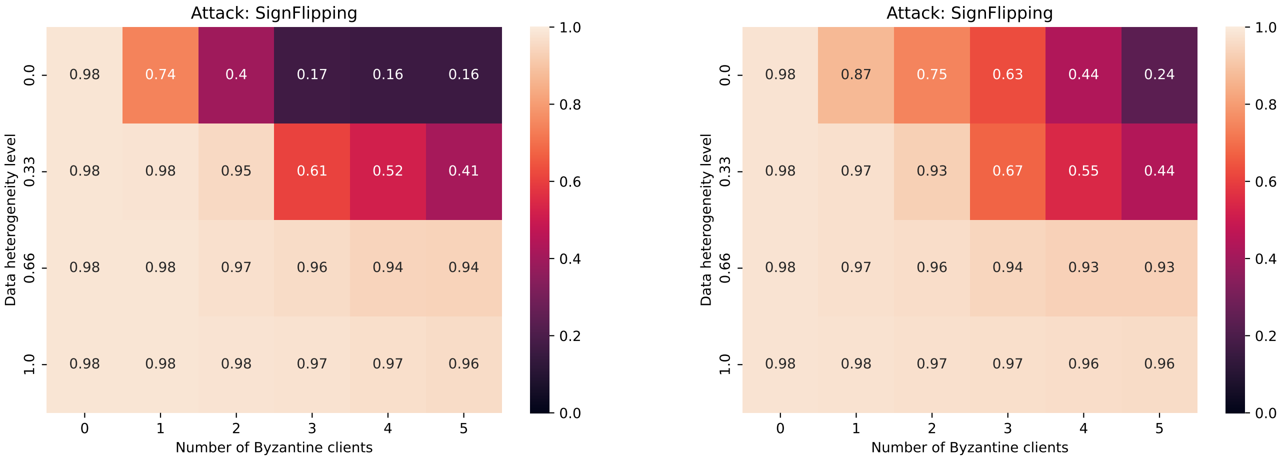
(a) CNN ReLU (No Clip)



(b) CNN Tanh

Figure 10: Centered Clipping under Optimal ALIE: CNN ReLU (No Clip) vs. CNN Tanh.

4.2.3 CC under Sign Flipping

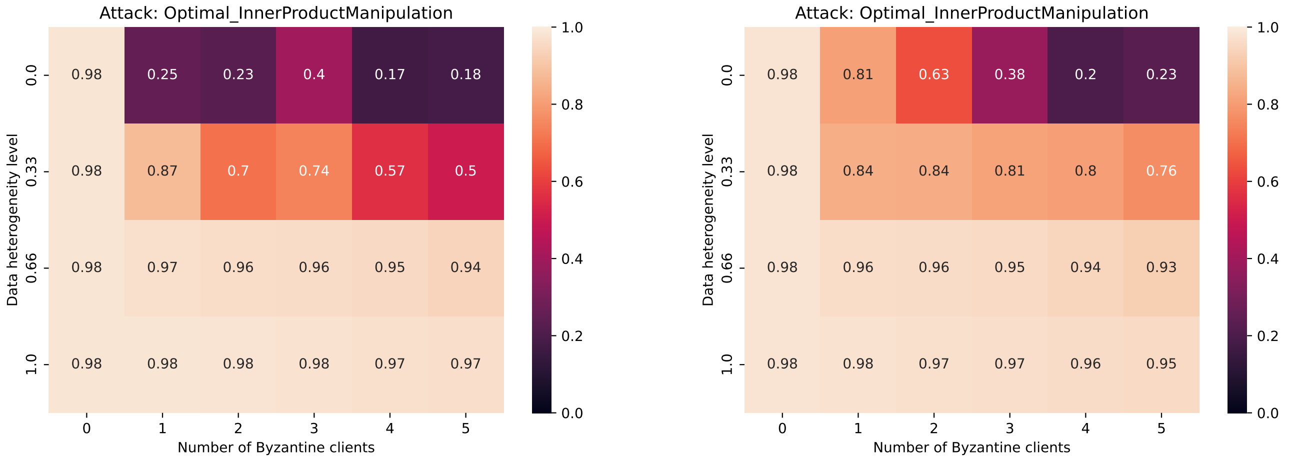


(a) CNN ReLU (No Clip)

(b) CNN Tanh

Figure 11: Centered Clipping under Sign Flipping: CNN ReLU (No Clip) vs. CNN Tanh.

4.2.4 CC under Optimal IPM



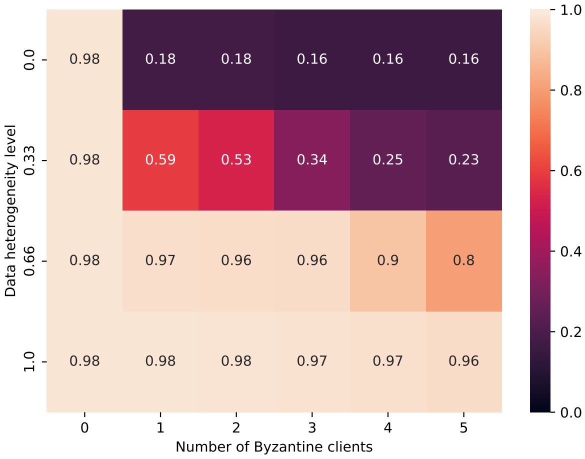
(a) CNN ReLU (No Clip)

(b) CNN Tanh

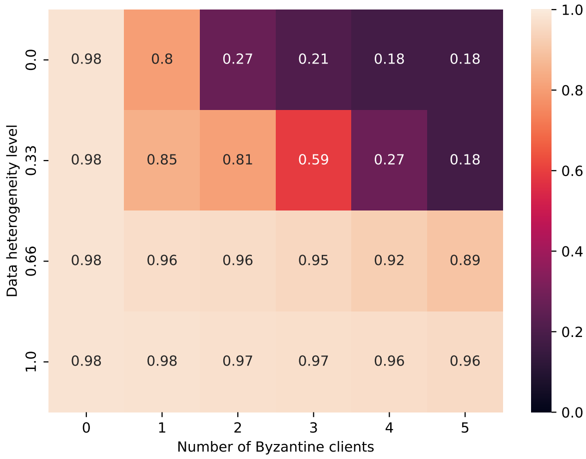
Figure 12: Centered Clipping under Optimal IPM: CNN ReLU (No Clip) vs. CNN Tanh.

4.3 Multi-Krum (MK) — NoClip vs Tanh

4.3.1 MK under Worst-Case



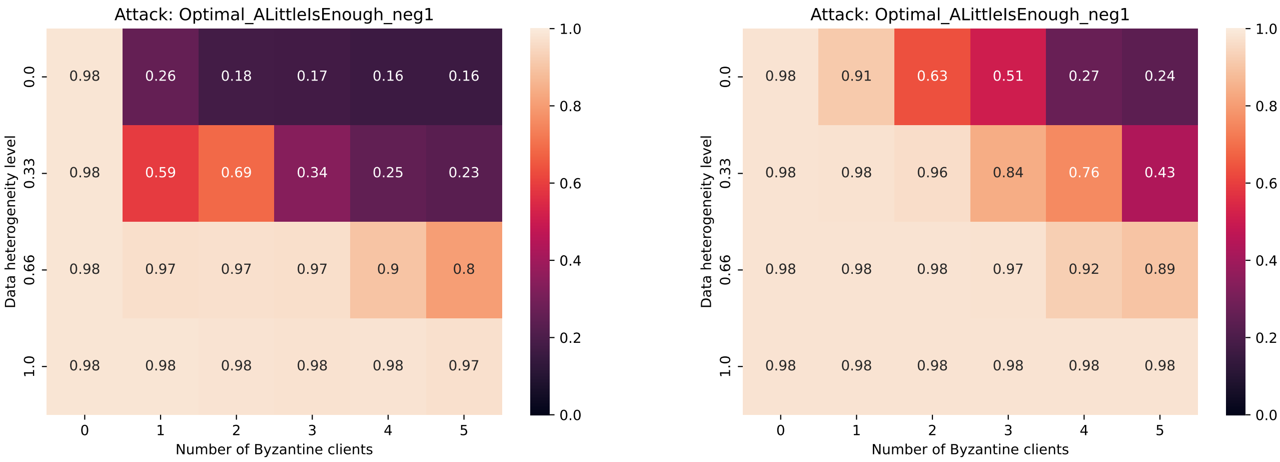
(a) CNN ReLU (No Clip)



(b) CNN Tanh

Figure 13: Multi-Krum under Worst-Case: CNN ReLU (No Clip) vs. CNN Tanh.

4.3.2 MK under Optimal ALIE

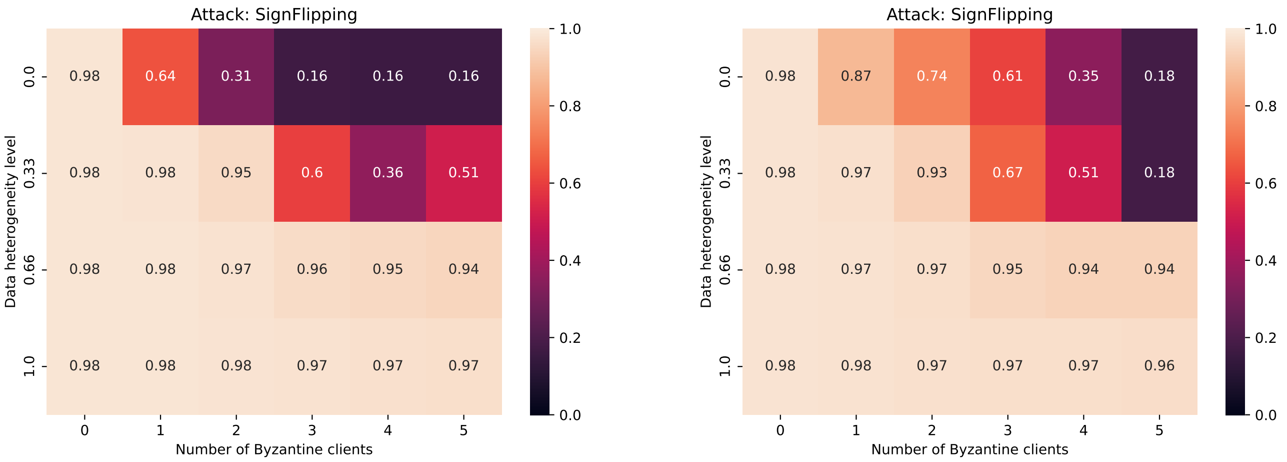


(a) CNN ReLU (No Clip)

(b) CNN Tanh

Figure 14: Multi-Krum under Optimal ALIE: CNN ReLU (No Clip) vs. CNN Tanh.

4.3.3 MK under Sign Flipping

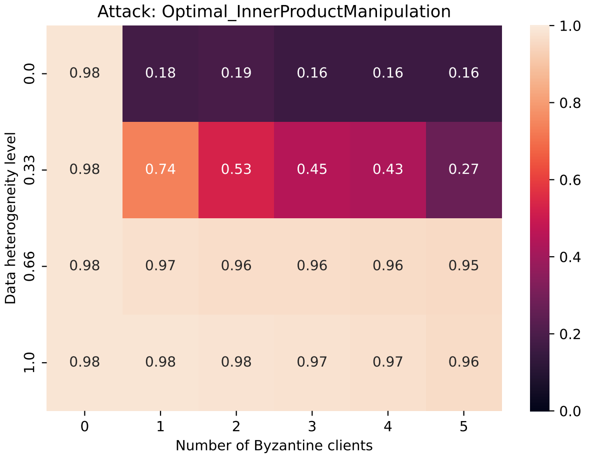


(a) CNN ReLU (No Clip)

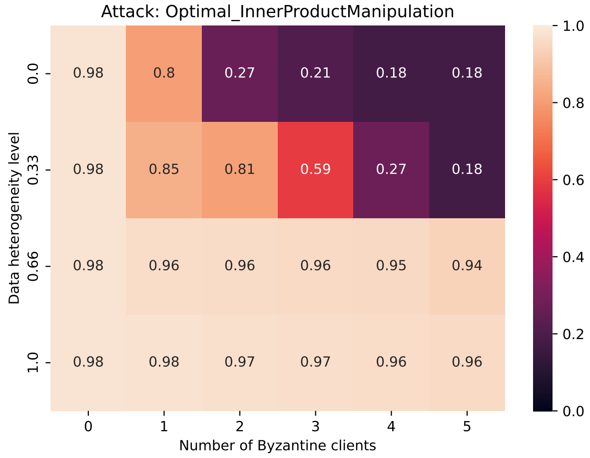
(b) CNN Tanh

Figure 15: Multi-Krum under Sign Flipping: CNN ReLU (No Clip) vs. CNN Tanh.

4.3.4 MK under Optimal IPM



(a) CNN ReLU (No Clip)

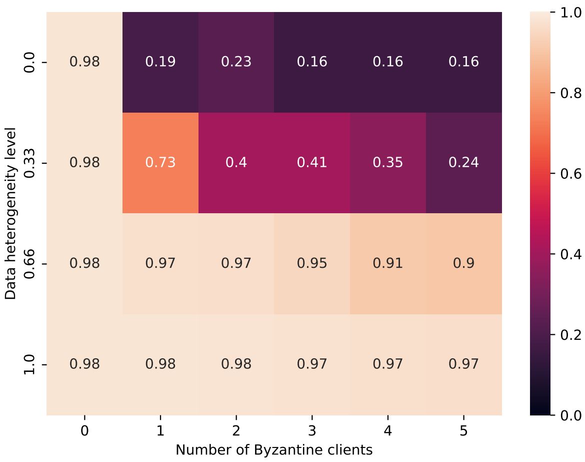


(b) CNN Tanh

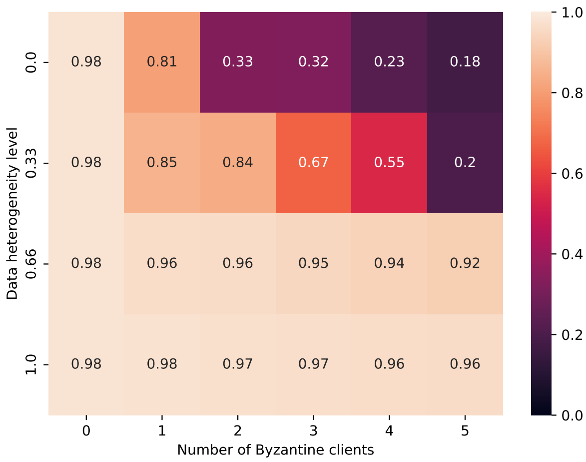
Figure 16: Multi-Krum under Optimal IPM: CNN ReLU (No Clip) vs. CNN Tanh.

4.4 Trimmed Mean (TM) — NoClip vs Tanh

4.4.1 TM under Worst-Case



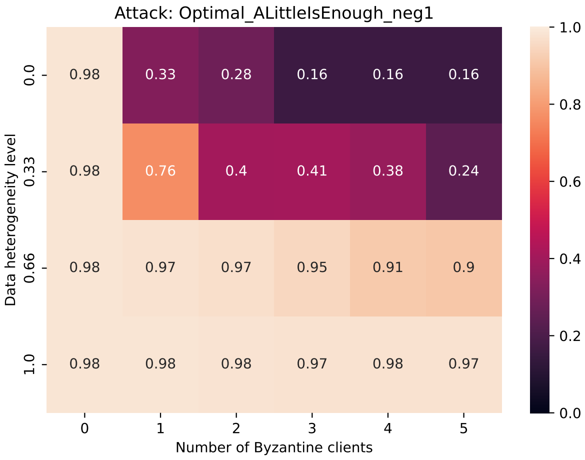
(a) CNN ReLU (No Clip)



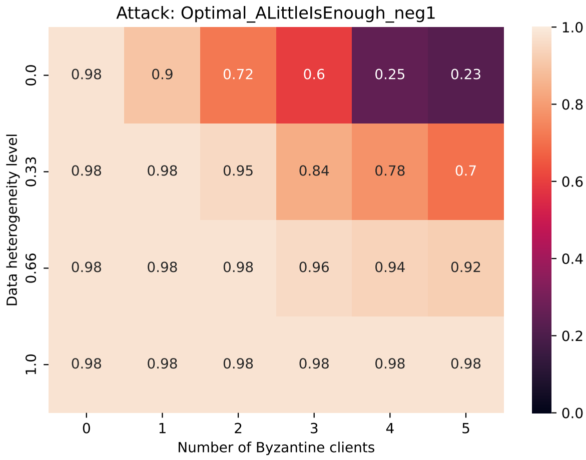
(b) CNN Tanh

Figure 17: Trimmed Mean under Worst-Case: CNN ReLU (No Clip) vs. CNN Tanh.

4.4.2 TM under Optimal ALIE



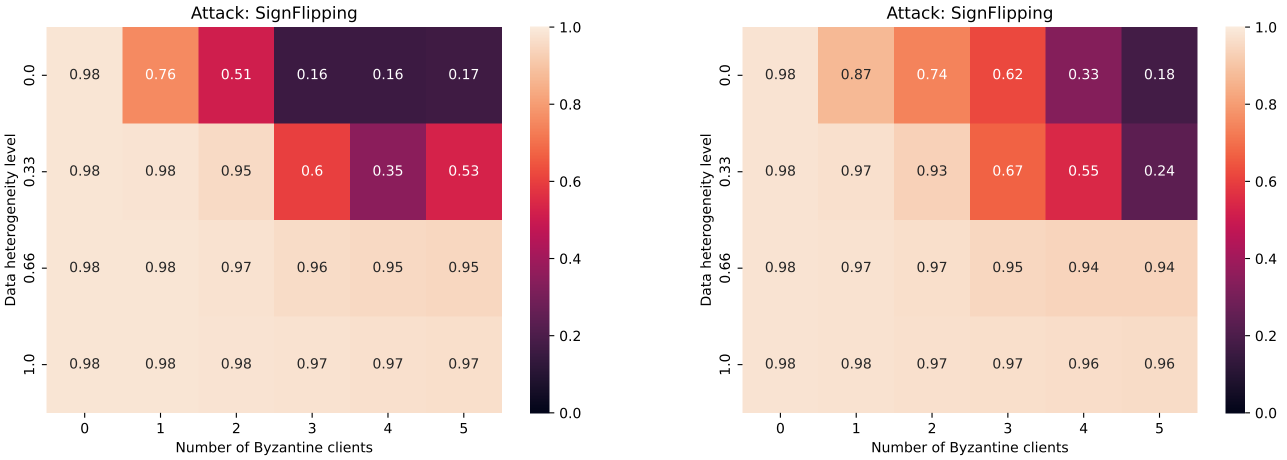
(a) CNN ReLU (No Clip)



(b) CNN Tanh

Figure 18: Trimmed Mean under Optimal ALIE: CNN ReLU (No Clip) vs. CNN Tanh.

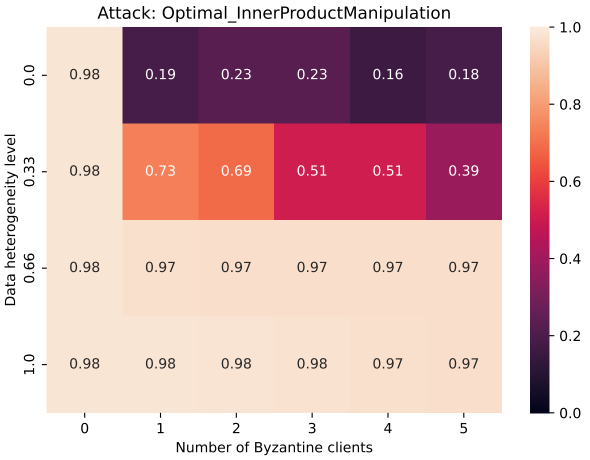
4.4.3 TM under Sign Flipping



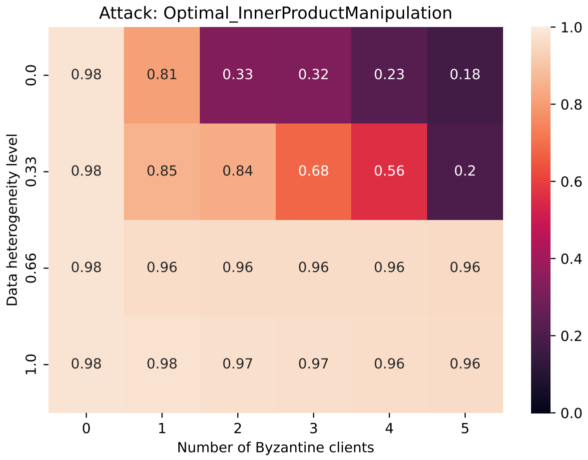
(a) CNN ReLU (No Clip) (b) CNN Tanh

Figure 19: Trimmed Mean under Sign Flipping: CNN ReLU (No Clip) vs. CNN Tanh.

4.4.4 TM under Optimal IPM



(a) CNN ReLU (No Clip)



(b) CNN Tanh

Figure 20: Trimmed Mean under Optimal IPM: CNN ReLU (No Clip) vs. CNN Tanh.